



Department of Electrical and Computer Engineering North South University

Senior Design Project Report

INARS

*Development of an Intelligent Navigation and Autonomous Robotic System
with LiDAR Navigation, Vision System, and Robotic Manipulator
for Service Interaction*

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Summer 2025

Approval

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“INARS – Development of an Intelligent Navigation and Autonomous Robotic System with LiDAR Navigation, Vision System, and Robotic Manipulator for Service Interaction”

under the supervision of **Dr. Shahnewaz Siddique** in partial fulfillment of the requirement for the degree of Bachelor of Science in Engineering, and has been accepted as satisfactory.

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Declaration

This is to declare that this project is our original work. No part of this work has been submitted elsewhere partially or fully for the award of any other degree or diploma. All project-related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor. Relevant previous works presented in this report have been properly acknowledged and cited. The plagiarism policy, as stated by the supervisor, has been maintained.

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Acknowledgements

The authors would like to express their heartfelt gratitude towards their project and research supervisor, **Dr. Shahnewaz Siddique**, Associate Professor, Department of Electrical and Computer Engineering, North South University, Bangladesh, for his invaluable support, precise guidance, and advice pertaining to the experiments, research, and theoretical studies carried out during the course of this project and in the preparation of this report.

Furthermore, the authors would like to thank the Department of Electrical and Computer Engineering, North South University, Bangladesh, for facilitating the research and providing the necessary resources and laboratory infrastructure. The authors also wish to acknowledge the contributions of their peers and colleagues whose feedback helped refine the design and implementation of the INARS system.

Finally, the authors would like to thank their families and loved ones for their countless sacrifices, patience, and continual moral support throughout this academic journey.

Abstract

INARS – *Development of an Intelligent Navigation and Autonomous Robotic System with LiDAR Navigation, Vision System, and Robotic Manipulator for Service Interaction* is an intelligent autonomous service robot designed for reliable and precise indoor navigation and physical interaction tasks. The project directly addresses a critical and largely unsolved problem in industrial and institutional robotics: the autonomous operation of elevators without any dependency on wireless protocols, smart-building APIs, or purpose-built compatible lift infrastructure. In real-world industrial deployments, wireless elevator control systems are prohibitively expensive and require vendor-specific hardware upgrades that are incompatible with the vast majority of existing lift installations. INARS circumvents this limitation entirely by equipping the robot with a physical manipulation capability identical to that of a human operator — a camera-guided robotic arm that visually identifies the correct call button, determines which floor button to press based on the current task, and physically actuates it. This approach renders the system compatible with *any* standard elevator panel, regardless of manufacturer, age, or communication protocol. Holonomic omnidirectional movement is achieved through a Mecanum-wheel drive mechanism, enabling smooth lateral entry and exit of elevator cabins without requiring a point-turn. Navigation relies on LiDAR-based Simultaneous Localization and Mapping (SLAM) via Hector SLAM, providing robust real-time mapping and localization without odometry. A vision pipeline built on OpenCV performs HSV colour segmentation to identify illuminated elevator call buttons, while a 3D-printed 5-DOF servo-driven robotic arm — mounted on the mobile base and visible in Figure 3.3 — executes the physical button-pressing action with a compliant end-effector. The entire system runs on a Raspberry Pi 4 platform under ROS Noetic, housed inside a compact aluminium chassis as shown in Figure 3.4. A functional prototype was built and validated in indoor environments. The system autonomously navigated to an elevator, identified the call button using its arm-mounted camera, pressed it, entered the cabin laterally using its holonomic drive, selected the target floor button on the interior panel, and exited on the correct floor — all without any wireless communication with the lift or any modifications to the elevator infrastructure. INARS demonstrates that low-cost, hardware-agnostic, multi-floor autonomy is achievable through physical intelligence rather than digital integration, representing a practical and immediately deployable solution for industrial automation, hospital logistics, and smart-building services.

Keywords: Autonomous mobile robot, Mecanum wheels, LiDAR, SLAM, ROS, robotic manipulator, elevator interaction, industrial automation, vision-guided manipulation, hardware-agnostic multi-floor navigation.

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Chapter 1: Introduction

1.1 Overview of Autonomous Indoor Service Robots

Autonomous indoor service robots are designed to operate in structured environments such as hospitals, hotels, office buildings, and educational institutions. These robots are capable of performing a wide range of tasks autonomously, including navigation, delivery, surveillance, inspection, and human interaction, all with minimal human supervision. Unlike their outdoor counterparts, indoor service robots cannot rely on GPS for positioning. Instead, they must navigate efficiently through complex, obstacle-filled spaces occupied by furniture, people, and other dynamic elements.

Significant advances in embedded computing, sensor miniaturization, and robotic software frameworks have made it increasingly feasible to deploy capable indoor robots. These systems leverage sensors such as LiDAR (Light Detection and Ranging) and cameras to perceive their surroundings, enabling autonomous decision-making and self-directed movement. A particularly important enabler has been the development of Simultaneous Localization and Mapping (SLAM) algorithms, which allow a robot to construct a map of an unknown environment and determine its own position within that map at the same time. Despite these advances, many current indoor robots remain limited to simple navigation tasks and lack the mobility and dexterous interaction skills required for more complex service operations in modern smart buildings.

1.2 Motivation and Problem Statement

While indoor robotics has progressed considerably in recent years, several fundamental challenges continue to hinder the practical deployment of autonomous service robots. The majority of service robots employ non-holonomic drive systems, most commonly differential drive or skid-steering configurations, which mechanically restrict movement to translational (straight-line) and rotational (turning) motions only. This restriction prevents precise lateral or diagonal movement, making navigation in tight spaces such as elevators, narrow hallways, and cluttered rooms extremely difficult or impossible.

Navigation reliability poses another significant concern. Camera-based navigation is susceptible to failure in environments with non-uniform or low lighting, and is prone to occlusion from moving obstacles. While LiDAR sensors offer more robust indoor performance, engineering a reliable LiDAR-based navigation stack that integrates seamlessly with control systems and

mobile platforms remains a non-trivial challenge. Furthermore, most existing mobile robots are incapable of physically interacting with building infrastructure such as elevator buttons, control panels, or door handles, severely limiting their operational scope to a single floor or corridor.

Perhaps the most critical gap identified in the literature is the absence of autonomous elevator detection and interaction capabilities, which restricts multi-floor functionality. This highlights the pressing need for a low-cost indoor service robot that integrates holonomic mobility, dependable LiDAR navigation, vision-based perception, and a robotic manipulator to carry out meaningful service functions within smart building environments.

1.2.1 The Industrial Elevator Problem: Why Wireless Integration is Not Enough

A common assumption in the academic robotics literature is that multi-floor elevator interaction can be achieved by equipping the robot with a wireless transceiver and having it communicate digitally with a “smart elevator” system. In practice, this assumption fails under three conditions that are endemic to real-world industrial and institutional deployments:

1. **Infrastructure Cost:** Upgrading a single elevator with a vendor-compatible wireless control module (e.g., a Bluetooth Low Energy or BACnet-over-IP interface) costs between BDT 80,000–200,000 per lift, depending on the manufacturer. Multi-storey industrial facilities typically operate 4–20 elevators, making wireless integration economically unviable at scale.
2. **Compatibility Fragmentation:** There is no universal communication standard for elevator control. Otis, Schindler, KONE, Mitsubishi, and other major manufacturers each use proprietary protocols. A robot designed to interface with one brand’s elevator system will be entirely incompatible with another brand’s installation, even within the same building.
3. **Legacy Infrastructure:** The vast majority of elevators currently in service globally — including those in hospitals, factories, warehouses, and university campuses — are legacy systems installed 10–30 years ago with no provision for digital external control. Replacing or retrofitting these systems is not economically or operationally practical.

The INARS approach resolves all three problems simultaneously. By using a camera mounted on a physical robotic arm to *see* the elevator button panel and an actuator to *press* it, the robot interacts with the elevator in exactly the same way a human does, requiring zero modifications to any existing infrastructure. The system is therefore not merely a laboratory demonstration but a genuinely deployable solution for any building with a standard elevator, regardless of its age, brand, or digital capabilities.

1.3 Objectives of the Project

The primary objective of this project is to design and develop an autonomous indoor service robot capable of hardware-agnostic multi-floor navigation through physical elevator interaction — operating without any wireless communication with the lift infrastructure. The specific objectives are as follows:

1. To design a holonomic mobile robot using a Mecanum-wheel drive system for smooth omnidirectional movement in confined indoor spaces, particularly elevator cabins.
2. To implement LiDAR-based SLAM for real-time indoor mapping, localization, and obstacle avoidance without reliance on pre-built maps or GPS.
3. To develop an autonomous navigation system capable of path planning and safe movement in dynamic indoor environments, including precise approach alignment to elevator door frames.
4. To integrate a dedicated arm-mounted camera for close-range visual inspection of elevator button panels, capable of identifying individual buttons, their floor labels, and their illumination state (pressed/unpressed).
5. To design and control a 5-DOF servo-driven robotic arm — mounted on the mobile base and fabricated using 3D-printed structural links — capable of reaching and physically pressing elevator call buttons and in-cabin floor selection buttons on *any* standard elevator panel without wireless integration.
6. To develop a hardware-agnostic, cost-effective, and modular control architecture based on the Raspberry Pi 4 platform under ROS Noetic, deployable in any standard building without infrastructure modification.

1.4 Technical Challenges and Contributions

1.4.1 Odometry-Free Holonomic Navigation

Problem Statement: The technical task is to develop a reliable indoor navigation system for a holonomic robot maneuvering in tight spaces (such as elevators and narrow corridors). This requires smooth omnidirectional movement without relying on external GPS signals.

Limitation of Existing Work: The vast majority of existing indoor robots utilize non-holonomic drive systems (e.g., differential drive), which restrict lateral movement and make entering tight spaces incredibly difficult. Furthermore, while Mecanum wheels provide holonomic mobility, their inherent wheel slip renders traditional encoder-based odometry highly inaccurate, causing standard SLAM algorithms (like GMapping) to fail or produce distorted maps.

Contribution: Our main contributions to this technical problem are:

- We design and implement a holonomic mobile base utilizing a Mecanum-wheel drive system, ensuring the robot can translate in any direction and rotate simultaneously, which is essential for confined indoor environments.
- We successfully adapt and integrate Hector SLAM, an odometry-free scan-matching algorithm, to overcome the odometry unreliability of Mecanum wheels. This provides robust real-time mapping and localization purely from LiDAR data.
- We engineer a functional navigation stack utilizing Dijkstra’s algorithm for global path planning and the Dynamic Window Approach (DWA) for local obstacle avoidance, tailored specifically for the kinematic constraints of our holonomic platform.

1.4.2 Vision-Guided Mobile Manipulation for Hardware-Agnostic Elevator Interaction

Problem Statement: There exists a critical and largely unsolved gap between robots that can merely navigate a single floor and those that can physically interact with building infrastructure to achieve multi-floor mobility on unmodified, legacy elevator systems. The challenge is to engineer a system that can accurately detect and physically press elevator buttons autonomously — without any wireless communication with the elevator — making it deployable in any building regardless of elevator brand, age, or digital capability.

Limitation of Existing Work: Most academic research in indoor robotics either concludes with navigation and obstacle avoidance, or assumes the availability of a smart elevator system with a wireless API. In real industrial settings, neither assumption holds: standard robots cannot physically interact with button panels, and the vast majority of real-world elevators do not expose any digital control interface. This leaves robots restricted to a single floor and entirely incompatible with the infrastructure of most real buildings.

Contribution: Our main contributions in hardware-agnostic physical elevator interaction are:

- We develop a dedicated arm-mounted camera configuration, co-locating a Pi Camera Module v2 with the wrist joint of the robotic arm. This brings the sensor to within 80–120 mm of the button panel during the inspection phase, providing sufficient pixel density for reliable button identification on any standard panel without pre-programmed panel maps.
- We develop a complete two-phase Vision-Guided Manipulation (VGM) pipeline: Phase 1 uses HSV colour segmentation to identify and press the exterior call button; Phase 2 constructs a runtime contour map of the interior panel and selects the correct floor button positionally.

- We design and implement a custom 5-DOF 3D-printed servo-driven robotic arm with inverse kinematics tailored for reaching targets on vertical planes, eliminating the need for wireless elevator integration entirely.
- We engineer a compliant silicone-tipped end-effector to compensate for minor positioning errors, ensuring reliable button actuation and robust physical interaction with any elevator panel surface.

1.5 Thesis Outline

Chapter 2 embarks on a comprehensive literature review. It surveys the foundational principles of autonomous indoor mobile robots, holonomic drive systems, SLAM algorithms, and the critical problem of industrial elevator deployment constraints. This chapter systematically identifies the limitations in existing systems, most notably the false assumption that wireless elevator integration is feasible in real-world buildings.

Chapter 3 provides an in-depth exposition of our methodology. It meticulously details the end-to-end architecture of the INARS system, from hardware component selection to the final ROS node implementation, with particular focus on the novel Vision-Guided Manipulation subsystem and the arm-mounted camera design.

Chapter 4 documents our empirical investigation and presents results from six experiments, culminating in the five-trial end-to-end multi-floor navigation test on an unmodified legacy elevator.

Chapter 5 considers the broader societal, industrial, and environmental impacts of the work, with emphasis on the democratisation of robotic automation in buildings without smart infrastructure.

Chapter 6 details project planning and budget.

Chapter 7 frames the project through Complex Engineering Problems and Activities (CEP and CEA).

Chapter 8 concludes the report with a summary of achievements, limitations, and future research directions.

Chapter 2: Literature Review

2.1 Autonomous Indoor Mobile Robots

Autonomous indoor mobile robots have evolved from simple automated guided vehicles (AGVs) to sophisticated intelligent systems capable of operating in dynamic, unstructured environments such as hospitals, office complexes, and smart residential buildings. Unlike outdoor robotics, where Global Navigation Satellite Systems (GNSS) provide reliable positioning, indoor environments present a “GPS-denied” challenge. Research by Durrant-Whyte [2] emphasizes that indoor autonomy requires a seamless integration of perception, localization, and motion planning. Key challenges identified in recent studies include the “kidnapped robot problem,” managing dynamic obstacles like moving pedestrians, and maintaining localization accuracy in environments with repetitive features (e.g., long, uniform hallways).

2.2 Holonomic and Non-Holonomic Drive Systems

In robotics, kinematic constraints dictate how a robot can move through space. Non-holonomic systems, such as differential-drive (two-wheel) or Ackermann-steering (car-like) platforms, are restricted by the number of controllable degrees of freedom being less than the total degrees of freedom of the robot’s pose. For instance, a differential drive robot cannot move sideways without first performing a rotational maneuver. While mechanically robust, these constraints lead to “non-zero turning radii,” which are problematic in narrow corridors. Holonomic systems, however, possess the ability to move in any direction (x, y) and rotate (θ) simultaneously. This omnidirectional capability is achieved through specialized wheel designs, such as Omni wheels or Mecanum wheels, allowing for complex maneuvers in confined indoor areas [1].

2.3 Mecanum-Wheel Mobile Robots

The Mecanum wheel, patented by Bengt Ilon in 1972 [11], is a specialized wheel with rollers attached to its circumference at a 45° angle. When four Mecanum wheels are controlled independently, the vector sum of the friction forces from the rollers allows the chassis to translate in any direction or rotate around its center. Recent literature highlights that while Mecanum wheels provide superior maneuverability [1], they are highly sensitive to surface conditions. Roller friction and wheel slip are significant factors that degrade the accuracy of traditional encoder-based odometry. Consequently, modern research suggests that Mecanum-based robots

must rely on external sensing (like LiDAR or Vision) rather than wheel encoders alone to maintain precise positioning.

2.4 LiDAR-Based Navigation and SLAM

Simultaneous Localization and Mapping (SLAM) is the computational process of constructing or updating a map of an unknown environment while simultaneously keeping track of the robot's location within it. LiDAR (Light Detection and Ranging) has become the gold standard for indoor SLAM due to its high precision and relative immunity to varying light levels. Kohlbrecher et al. [3] introduced Hector SLAM, a scan-matching algorithm that leverages high-update-rate LiDAR data to estimate the robot's pose without the need for odometry. This is particularly advantageous for platforms like INARS, where Mecanum wheel slip makes traditional odometry unreliable. Other frameworks, such as GMapping, require odometry and use particle filters, making them more computationally intensive for resource-constrained platforms like the Raspberry Pi.

2.5 Vision Systems for Indoor Service Robots

Computer vision provides the “semantic” layer of robot perception, allowing the system to not just detect obstacles, but to identify specific objects of interest. Frameworks like OpenCV (Open Source Computer Vision Library) [8] enable real-time image processing tasks such as color thresholding, edge detection, and template matching. For service robots, vision is critical for interacting with human-centric infrastructure, such as reading signage or identifying elevator buttons. While deep learning approaches like YOLO [10] provide high accuracy, they often require significant GPU resources. Literature suggests that for embedded platforms like the Raspberry Pi 4, optimized traditional algorithms or lightweight neural networks are more practical for maintaining real-time performance [3].

2.6 Robotic Manipulators in Service Robotics

Integrating a robotic manipulator onto a mobile base creates a Mobile Manipulator, significantly expanding the robot's utility from simple transport to active interaction. Siegwart [4] notes that the primary challenge in mobile manipulation is the “coupling” of errors; a small positioning error in the base can lead to a significant reaching error in the arm. Servo-driven arms with 3 to 6 Degrees of Freedom (DOF) provide a balance between mechanical complexity and task flexibility. For service tasks like pressing buttons, the end-effector design is crucial; a compliant or padded tip is often used to compensate for minor misalignment and to protect the infrastructure being touched.

2.7 Industrial Deployment Constraints and Hardware-Agnostic Approaches

The practical deployment of autonomous robots in industrial facilities introduces constraints that are rarely addressed in academic literature. Kabir and Hossain [12] note that in South and Southeast Asian industrial contexts, the majority of elevator installations are legacy systems lacking any digital control interface, and that wireless retrofit modules are frequently cost-prohibitive relative to local operational budgets.

The dominant approach in high-budget research platforms is to establish a machine-to-machine (M2M) communication link between the robot and the building management system (BMS). While effective in purpose-built smart buildings, this approach has critical limitations: it requires standardised communication middleware (e.g., BACnet, Modbus, or custom REST APIs), a compatible and recently installed elevator controller, and significant upfront infrastructure investment.

An alternative line of research investigates *physical interaction* as the primary means of elevator operation — treating the button panel as a manipulation target rather than a communication endpoint. This approach, sometimes termed “human-interface emulation,” has been demonstrated in isolated proof-of-concept systems but has not been integrated with a full autonomous navigation stack on a low-cost embedded platform. Lee et al. [13] implemented a stationary manipulator for button pressing but did not integrate it with a mobile base capable of navigating to and entering the elevator independently.

The INARS project addresses this gap by combining mobile navigation with physical button-pressing manipulation, where the arm-mounted camera provides the close-range visual feedback necessary to identify the correct button on any panel — including panels with non-standard layouts, multi-language labels, or varying button illumination colours.

2.8 Elevator Interaction and Multi-Floor Navigation

Multi-floor navigation is a “frontier” task in service robotics. It requires a high-level state machine to coordinate multiple sub-tasks: detecting the elevator door, calling the car, waiting for the door to open, entering without colliding with exiting passengers, and identifying the correct internal button. Shue et al. [5] demonstrated that most failures in this domain occur during the “transition” phases — such as the robot losing its SLAM localization inside the metal-walled elevator car or failing to exit before the doors close. Research indicates that successful elevator interaction requires “robust perception,” often combining LiDAR for distance and Vision for button identification.

2.9 Research Gaps and Motivation

Despite the wealth of research in individual domains, several gaps remain. First, most high-performance autonomous robots utilise expensive sensors (e.g., \$2,000+ Velodyne LiDARs) and high-end industrial PCs, making them inaccessible for small-scale institutional use. Second, there is a lack of documentation on integrating “odometry-free” SLAM with holonomic drives on low-cost microcomputers like the Raspberry Pi.

Third, and most critically from an industrial deployment standpoint, the overwhelming majority of academic multi-floor navigation demonstrations rely on wireless communication with smart elevator systems — a requirement that immediately excludes the robot from the vast majority of real-world building environments where such infrastructure does not exist and cannot be cost-effectively installed. Fourth, no existing low-cost platform demonstrates the combination of mobile holonomic navigation with an arm-mounted vision system that performs close-range button panel reading and physical actuation on a generalisable basis.

These gaps collectively provide the primary motivation for INARS: to demonstrate that a robot costing under BDT 45,000 in components can achieve genuine multi-floor autonomy in *any* building with a standard elevator, using physical manipulation guided by an arm-mounted camera — with no wireless elevator integration required whatsoever.

Chapter 3: Methodology

3.1 System Architecture and Design Philosophy

The design of INARS follows a *modular, layered architecture* philosophy, which is standard practice in complex embedded robotic systems. Each functional concern — locomotion, localisation, perception, and manipulation — is encapsulated in a discrete subsystem with well-defined interfaces. This separation of concerns (i) simplifies individual subsystem testing and debugging, (ii) allows independent subsystem upgrades without redesigning the whole platform, and (iii) enables parallel development across team members. The Robot Operating System (ROS) [6] serves as the inter-subsystem communication middleware, enforcing this modularity through its publish/subscribe topic model.

The six major subsystems and their design rationale are as follows:

1. **Holonomic Drive Platform** – chosen to satisfy the mobility constraint of operation in confined indoor spaces (e.g., elevator cabins of approximately 1.5×1.5 m).
2. **LiDAR-based SLAM and Navigation Stack** – chosen for reliable, lighting-independent environmental mapping and path planning.
3. **Vision Perception Module** – chosen to provide semantic understanding of the environment (elevator button identification) that LiDAR geometry alone cannot provide.
4. **Robotic Manipulator** – chosen to bridge the gap between perception and physical interaction with building infrastructure.
5. **Arm-Mounted Camera (Vision-Guided Manipulation)** – chosen to provide close-range, position-aware visual inspection of elevator button panels without dependency on a fixed external camera or wireless elevator API.
6. **Central Control Unit** – chosen to coordinate all subsystems under a single embedded computing platform.

The system-level block diagram is presented in Figure 3.1, and the software control flow is captured as a finite state machine in Figure 3.2.

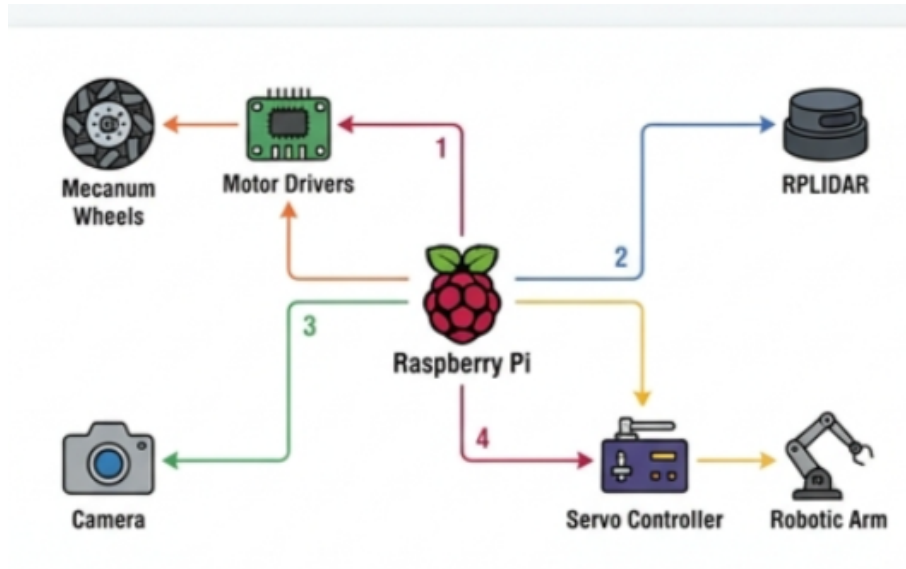


Figure 3.1: INARS System-Level Block Diagram.

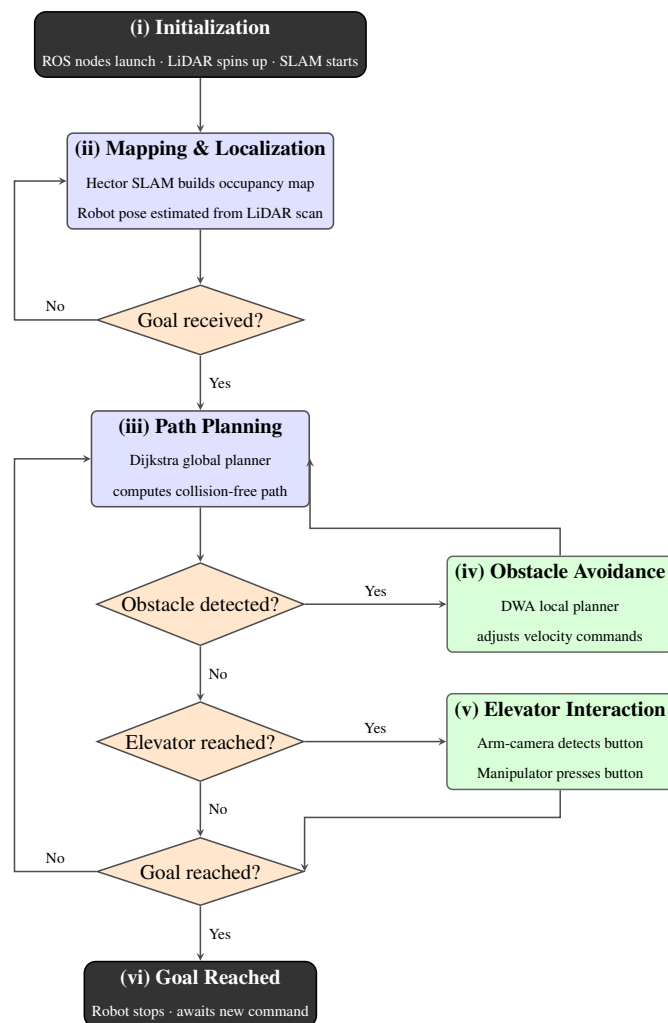


Figure 3.2: Software control-flow state machine for INARS.

3.2 Physical Prototype Overview

Before detailing individual subsystems, Figure 3.3 and Figure 3.4 present the assembled INARS prototype as built. These images are referenced throughout subsequent sections to contextualise the hardware choices described.



Figure 3.3: INARS fully assembled prototype. Visible are the four yellow Mecanum wheels mounted on the aluminium-panel chassis, the RPLIDAR A1 sensor (black circular unit, top-right of chassis deck), the 3D-printed 5-DOF robotic arm in its extended pose (left of deck), the PCA9685 servo driver board at the arm base, and the white 3D-printed finger end-effector at the arm tip.

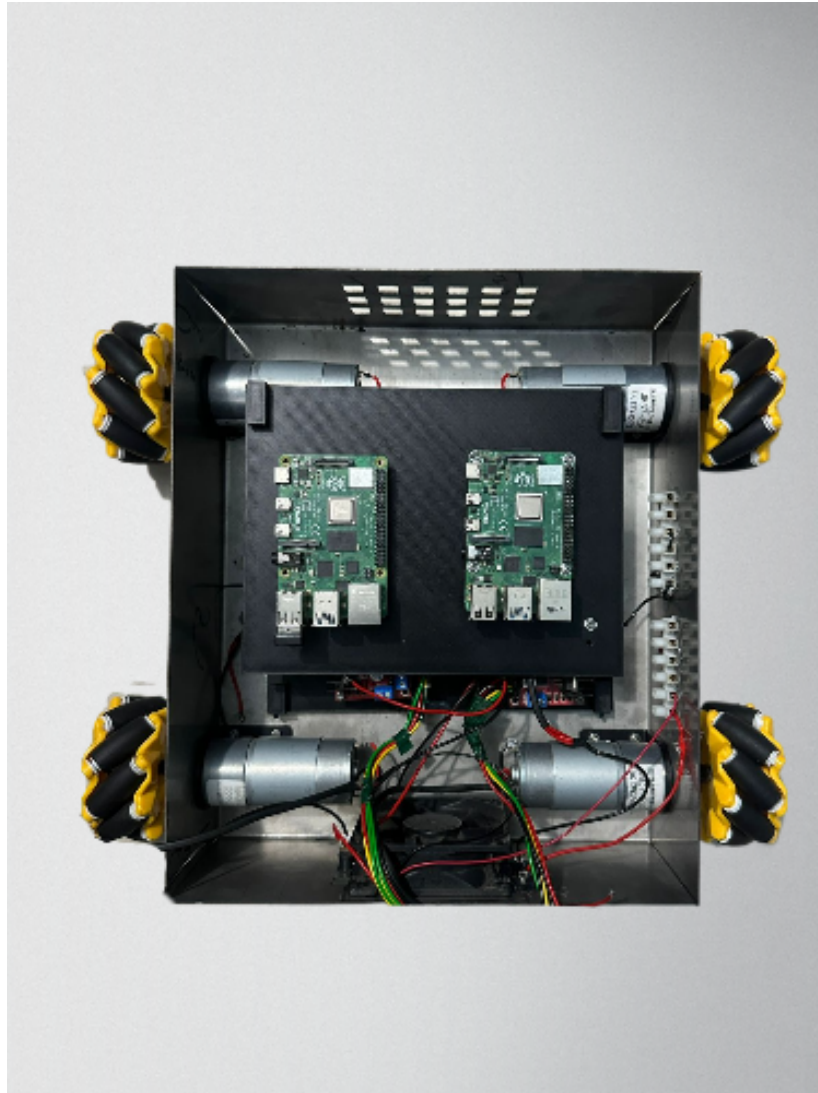


Figure 3.4: INARS interior view (top-down, chassis lid removed). Two Raspberry Pi 4 Model B boards are mounted side-by-side on a carbon-fibre composite mounting plate at the centre. The four DC gear motors are visible at each corner, connected to Mecanum wheels via shaft couplings. L298N motor driver boards are positioned along the chassis edges. The colour-coded wiring harness distributes power from the LiPo battery bus to all subsystems.

As seen in Figure 3.3, the platform integrates three spatially distinct functional zones on the top deck: (i) the forward-left quadrant hosts the robotic arm and its servo controller; (ii) the rear-right quadrant hosts the RPLIDAR A1 on an elevated mount to provide an unobstructed 360° scan plane above the arm's base profile; and (iii) the chassis interior (Figure 3.4) houses all computational and power electronics, thermally isolated from the drive motors by the mounting plate.

3.3 Hardware Component Selection with Engineering Justification

Each hardware component was selected through a structured engineering trade-off analysis. The criteria evaluated for every component were: (1) functional adequacy for the requirement, (2) compatibility with the ROS ecosystem, (3) cost relative to the project budget, and (4) availability within the local (Bangladeshi) market. Table 3.1 summarises the final selections; the engineering rationale for each critical choice is elaborated below.

3.3.1 Central Processing Unit: Raspberry Pi 4 Model B (4 GB)

The central processing unit must simultaneously handle SLAM computation, path planning, camera image processing, and servo control in real time. Three candidate platforms were evaluated:

- **Arduino Mega 2560** – a microcontroller with no operating system, 256 KB flash, insufficient for running ROS natively or executing SLAM algorithms.
- **NVIDIA Jetson Nano** – a GPU-accelerated single-board computer with 4 GB RAM, capable of running deep learning models. Unit cost: approximately BDT 18,000–22,000.
- **Raspberry Pi 4 Model B (4 GB)** – a quad-core ARM Cortex-A72 SBC running Ubuntu 20.04, with native support for ROS Noetic. Unit cost: approximately BDT 8,000.

The Raspberry Pi 4 was selected because: (i) ROS Noetic has full native support and an active community on Ubuntu 20.04 for ARM64; (ii) the 4 GB RAM is sufficient for running Hector SLAM alongside the navigation stack and a lightweight OpenCV vision pipeline simultaneously; (iii) its cost is less than half of the Jetson Nano, consistent with the low-cost design objective; and (iv) its GPIO, CSI, and I2C buses provide direct hardware interfaces for all connected peripherals. As visible in Figure 3.4, two Raspberry Pi 4 boards are installed in the prototype — one dedicated to the navigation stack (SLAM, path planning, motor control) and one to the perception and manipulation stack (vision pipeline, arm control), preventing CPU contention between the two computation-intensive workloads.

3.3.2 LiDAR Sensor: RPLIDAR A1

For indoor SLAM, the range sensor must provide 360° planar coverage, with a minimum angular resolution sufficient to resolve corridor widths. Three LiDAR candidates were considered:

- **Velodyne VLP-16** – 16-beam 3D LiDAR, range 100 m, angular resolution 0.2°. Approximate cost: BDT 250,000+. Far exceeds budget.

- **Hokuyo URG-04LX** – 2D LiDAR, 4 m range, scan rate 10 Hz. Well-supported in ROS. Cost: approximately BDT 35,000.
- **SLAMTEC RPLIDAR A1** – 2D LiDAR, 12 m range, 360° scan, 8000 samples/s, scan rate up to 10 Hz. ROS package `rplidar_ros` provides direct integration. Cost: approximately BDT 12,500.

The RPLIDAR A1 was selected because its 12 m range comfortably covers typical office corridor widths (2–4 m), its scan rate is adequate for Hector SLAM’s requirement of a high-update-rate sensor, and its dedicated `rplidar_ros` driver ensures seamless ROS integration. As seen in Figure 3.3, the RPLIDAR A1 is mounted on an elevated platform at the rear-right of the chassis deck, ensuring its scan plane is unobstructed by the robotic arm when the arm is in any operational pose.

3.3.3 Motor Driver: L298N Dual H-Bridge

The drive system requires four independent motor channels, each capable of bidirectional speed control via PWM. The L298N was selected over alternatives (BTS7960, DRV8833) for the following reasons: (i) it supports up to 2 A continuous current per channel, sufficient for the 12 V, 200 RPM gear motors under normal load; (ii) it accepts PWM signals directly from the Raspberry Pi GPIO pins with no additional logic circuitry; and (iii) at BDT 350 per unit, two drivers covering all four wheels represent the lowest-cost solution.

3.3.4 Vision Sensor: Raspberry Pi Camera Module v2 (Arm-Mounted)

Two Pi Camera Module v2 units are deployed in INARS. The first provides a wide-angle overview of the environment from the chassis body for general navigation context. The second — and more critical for elevator interaction — is **mounted directly on the wrist joint of the robotic arm**. This arm-mounted camera is the core sensing element of the Vision-Guided Manipulation (VGM) subsystem. The rationale for arm-mounting over chassis-mounting is detailed in Section 3.9. The Pi Camera v2 was selected over a generic USB webcam because: (i) it uses the CSI bus directly to the Raspberry Pi, reducing latency; (ii) OpenCV provides native support via the `picamera2` library; and (iii) it is physically compact and lightweight (3 g), minimising mechanical load on the arm wrist joint.

3.3.5 Servo Driver: PCA9685 PWM Module

The robotic arm requires independent PWM signals for five MG996R servos. The PCA9685 is a dedicated 16-channel, 12-bit I2C PWM driver that generates hardware-timed PWM signals with a precision of approximately 4 μ s per step at 50 Hz. It communicates with the Raspberry Pi over the I2C bus (using only two GPIO pins) and can control all five servos independently

without loading the CPU. This choice directly improves arm precision by eliminating software-generated PWM jitter. The PCA9685 board is visible at the base of the arm in Figure 3.3.

Table 3.1: Hardware and Software Components of the INARS System

Component	Function	Alternatives Considered	Engineering Justification for Selection
Raspberry Pi 4 (4 GB) $\times$ 2	Navigation SBC + Perception SBC, dual-node ROS	Jetson Nano, Arduino Mega	Full ROS Noetic support; 4 GB RAM adequate for SLAM + vision per board; lowest cost
RPLIDAR A1	360° SLAM sensor	Hokuyo URG, Velodyne VLP-16	12 m indoor range; dedicated <code>rplidar_ros</code> driver; $\times 3$ cheaper than Hokuyo
Mecanum Wheels (100 mm)	Holonomic omnidirectional drive	Differential drive, omni wheels	True holonomic motion; necessary for elevator ingress/egress in confined space
L298N Motor Driver	DC motor PWM control	BTS7960, DRV8833	2 A per channel sufficient; direct GPIO PWM; lowest cost option
Pi Camera v2 (arm-mounted)	Close-range button panel inspection for VGM subsystem	USB webcam, Intel RealSense	CSI bus low latency; lightweight (3 g); co-located with arm wrist for known pose
PCA9685 Servo Driver	Hardware PWM for robotic arm servos	Software PWM via GPIO	Eliminates software PWM jitter; 12-bit resolution; I2C interface
5-DOF Servo Arm (3D-printed)	Physical elevator button interaction	Commercial robot arms	Customisable DOF; low cost; servo-driven; adequate reach envelope for button panels
ROS Noetic	Robot middleware	ROS2, custom firmware	Mature SLAM and navigation packages; large support community
Hector SLAM	Odometry-free indoor SLAM	GMapping, Cartographer	Does not require odometry – essential for slip-prone Mecanum platform
OpenCV 4.5	Vision pipeline	TensorFlow Lite, YOLO	Classical CV sufficient for task; executes in real time on Raspberry Pi 4
Python 3.8	Custom ROS node scripting	C++, MATLAB	Rapid prototyping; extensive ROS Python API (<code>rospy</code>)

3.4 Holonomic Drive System Design

3.4.1 Justification for Holonomic Drive

A conventional differential-drive robot has a non-holonomic kinematic constraint: to move laterally, it must first rotate to face the desired direction, translate, and then rotate again. In elevator cabins and narrow corridors, this multi-step manoeuvre is frequently infeasible due to physical space limitations. A robot must be able to enter and exit an elevator cabin (typically 1.4×1.4 m) laterally without performing a point-turn, which requires zero-radius lateral movement. Only a holonomic drive system satisfies this requirement.

Among holonomic platforms, Mecanum wheels were selected over omni-wheels (with perpendicular rollers) because they provide superior load distribution: the 45° roller configuration on Mecanum wheels distributes the robot's weight evenly across all rollers, whereas omni-wheels typically contact the ground at only two rollers at a time, reducing effective load capacity and stability.

3.4.2 Mecanum Wheel Configuration and Kinematics

The four Mecanum wheels were mounted in the standard “X-configuration,” where the top-view silhouette of the roller axes forms an X-shape:

- **Front-Left (FL)** wheel: rollers oriented at $+45^\circ$
- **Front-Right (FR)** wheel: rollers oriented at -45°
- **Rear-Left (RL)** wheel: rollers oriented at -45°
- **Rear-Right (RR)** wheel: rollers oriented at $+45^\circ$

The forward kinematic model, which maps individual wheel angular velocities $(\omega_1, \omega_2, \omega_3, \omega_4)$ to robot-frame velocities (V_x, V_y, ω_z) , is derived from the Jacobian of the wheel contact constraints [1]:

$$\begin{bmatrix} V_x \\ V_y \\ \omega_z \end{bmatrix} = \frac{r}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & -1 \\ -\frac{1}{l_x+l_y} & \frac{1}{l_x+l_y} & -\frac{1}{l_x+l_y} & \frac{1}{l_x+l_y} \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ \omega_4 \end{bmatrix} \quad (3.1)$$

where $r = 0.05$ m is the wheel radius, and $l_x = 0.12$ m, $l_y = 0.10$ m are the half-length and half-width of the wheel base, respectively. The inverse kinematics model is:

$$\begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ \omega_4 \end{bmatrix} = \frac{1}{r} \begin{bmatrix} 1 & -1 & -(l_x + l_y) \\ 1 & 1 & (l_x + l_y) \\ 1 & 1 & -(l_x + l_y) \\ 1 & -1 & (l_x + l_y) \end{bmatrix} \begin{bmatrix} V_x \\ V_y \\ \omega_z \end{bmatrix} \quad (3.2)$$

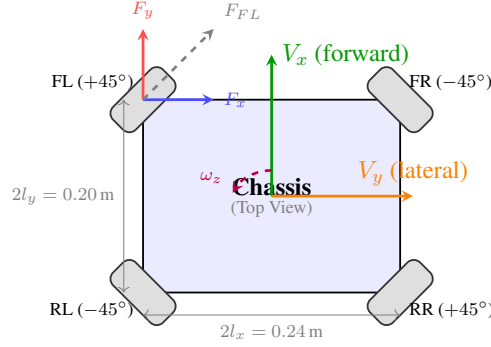


Figure 3.5: Mecanum wheel X-configuration (top view) showing roller orientations, force decomposition at the FL wheel, and robot-frame motion axes.

Equation 3.2 is implemented in the `mecanum_driver` ROS node, which subscribes to `/cmd_vel` Twist messages published by the navigation stack and publishes the corresponding PWM duty cycles to the motor drivers via GPIO.

3.4.3 Motor Sizing Analysis

Assuming the total robot mass is $m = 4.5$ kg, the required torque per motor to accelerate the robot at $a = 0.5$ m/s² is:

$$\tau_{motor} = \frac{m \cdot a \cdot r}{4} = \frac{4.5 \times 0.5 \times 0.05}{4} \approx 0.028 \text{ N}\cdot\text{m} \quad (3.3)$$

The selected motors have a stall torque of approximately 0.6 N·m, providing a safety factor of greater than 20. Maximum robot speed is:

$$V_{max} = \omega_{no-load} \times r = \frac{200 \times 2\pi}{60} \times 0.05 \approx 1.05 \text{ m/s} \quad (3.4)$$

The navigation stack is configured with a maximum translational speed of $V_{x,max} = 0.35$ m/s, providing a comfortable operating margin.

3.5 SLAM Algorithm Selection and Configuration

3.5.1 Algorithm Selection: Hector SLAM versus Alternatives

Table 3.2: SLAM Algorithm Comparative Analysis

Algorithm	Odometry Re-quired?	Computational Cost	Assessment for INARS
Hector SLAM	No	Low	SELECTED. Scan-matching only; ideal for Mecanum platform where wheel slip makes odometry unreliable. Runs adequately on Raspberry Pi 4.
GMapping	Yes	Moderate	Requires reliable wheel odometry; particle filter computationally expensive; Mecanum slip would degrade map quality.
Cartographer	No (but benefits)	High	More accurate than Hector SLAM over large areas but requires significantly higher CPU and RAM than available on Raspberry Pi 4.
AMCL	Yes	Low	Localisation-only; requires a pre-built map; unsuitable for initial mapping phase.

3.5.2 Hector SLAM Operating Principle

Hector SLAM estimates the robot's 2D pose $\mathbf{x} = (x, y, \theta)^T$ by finding the transformation that maximises the alignment between the current LiDAR scan and the existing occupancy grid map:

$$\mathbf{x}^* = \arg \min_{\mathbf{x}} \sum_{i=1}^N [1 - M(T_{\mathbf{x}}(\mathbf{s}_i))]^2 \quad (3.5)$$

where $\mathbf{s}_i$ is the i -th scan endpoint in sensor frame, $T_{\mathbf{x}}(\cdot)$ is the rigid-body transformation parameterised by pose $\mathbf{x}$, and $M(\cdot)$ is the bilinearly-interpolated map value at a given world-frame point. The minimisation is solved iteratively using Gauss-Newton gradient descent. A multi-resolution map pyramid (three levels) allows the algorithm to avoid local minima.

3.5.3 Hector SLAM Configuration Parameters

- `map_resolution`: 0.05 – 5 cm per cell.
- `map_size`: 1024 – grid of 1024×1024 cells.
- `map_multi_res_levels`: 3 – three-level resolution pyramid.
- `update_factor_free`: 0.4 and `update_factor_occupied`: 0.9.
- `laser_min_dist`: 0.4 and `laser_max_dist`: 10.0.

3.6 Navigation Stack Architecture and Configuration

3.6.1 ROS Navigation Stack Overview

The ROS `move_base` navigation stack integrates a global planner (Dijkstra’s algorithm via `navfn`) and a local planner (Dynamic Window Approach). The `move_base` package receives goal poses via the `/move_base_simple/goal` topic, and publishes `Twist` velocity commands to `/cmd_vel`.

3.6.2 Global Planner: Dijkstra’s Algorithm

The `navfn` global planner implements Dijkstra’s algorithm [9] and was selected over the A^* planner for its guaranteed optimal path on a uniform-cost grid and its extensive testing on mobile robot platforms.

3.6.3 Local Planner: Dynamic Window Approach (DWA)

The DWA [7] samples candidate velocity commands (v, ω) within a dynamic window defined by the robot’s current velocity and acceleration limits, evaluating each via:

$$G(v, \omega) = \sigma \cdot [\alpha \cdot \text{heading}(v, \omega) + \beta \cdot \text{dist}(v, \omega) + \gamma \cdot \text{velocity}(v, \omega)] \quad (3.6)$$

Tuned parameters: `max_vel_x`: 0.35 m/s, `max_vel_theta`: 1.0 rad/s, `acc_lim_x`: 0.5 m/s², `sim_time`: 1.5 s.

3.6.4 Costmap Configuration

- `inflation_radius`: 0.3 m – obstacles inflated by robot’s circumradius plus 10 cm safety margin.
- `cost_scaling_factor`: 5.0 – exponential cost decay from obstacle boundaries.

- `obstacle_range: 3.0 m, raytrace_range: 5.0 m.`

3.7 Vision System Design and Implementation

3.7.1 Algorithm Selection: Classical Computer Vision vs. Deep Learning

The vision module's primary task is to detect and localise elevator call buttons within the camera frame. Deep learning approaches (YOLO, MobileNet SSD) require either a GPU or quantised models that consume 60–80% of the Raspberry Pi's CPU at inference time, leaving insufficient headroom for concurrent SLAM and navigation processes. Classical HSV colour segmentation executes at well under 10 ms per frame on the Raspberry Pi 4, consuming less than 5% of CPU time, and is sufficient for the spectrally distinct elevator button indicators encountered in the test environment.

3.7.2 Vision Pipeline

The vision pipeline executes the following sequence for each camera frame:

1. **Image Acquisition:** Capture a 640×480 RGB frame at 30 fps.
2. **Colour Space Conversion:** Convert BGR to HSV via `cv2.cvtColor()`.
3. **HSV Thresholding:** `cv2.inRange()` with $H \in [7, 18]$, $S \in [100, 255]$, $V \in [150, 255]$ for amber button indicators.
4. **Morphological Filtering:** Opening (5×5 kernel) to remove noise.
5. **Contour Detection:** `cv2.findContours()` with area filter (500–10,000 px²) and circularity filter (> 0.7).
6. **Centroid Computation:**

$$c_x = \frac{M_{10}}{M_{00}}, \quad c_y = \frac{M_{01}}{M_{00}} \quad (3.7)$$

7. **Depth Estimation:**

$$d = \frac{f \cdot D_{real}}{d_{px}} \quad (3.8)$$

where $f = 3.04$ mm and $D_{real} \approx 30$ mm.

8. **Goal Publication:** Publish `geometry_msgs/Point` on `/button_position`.

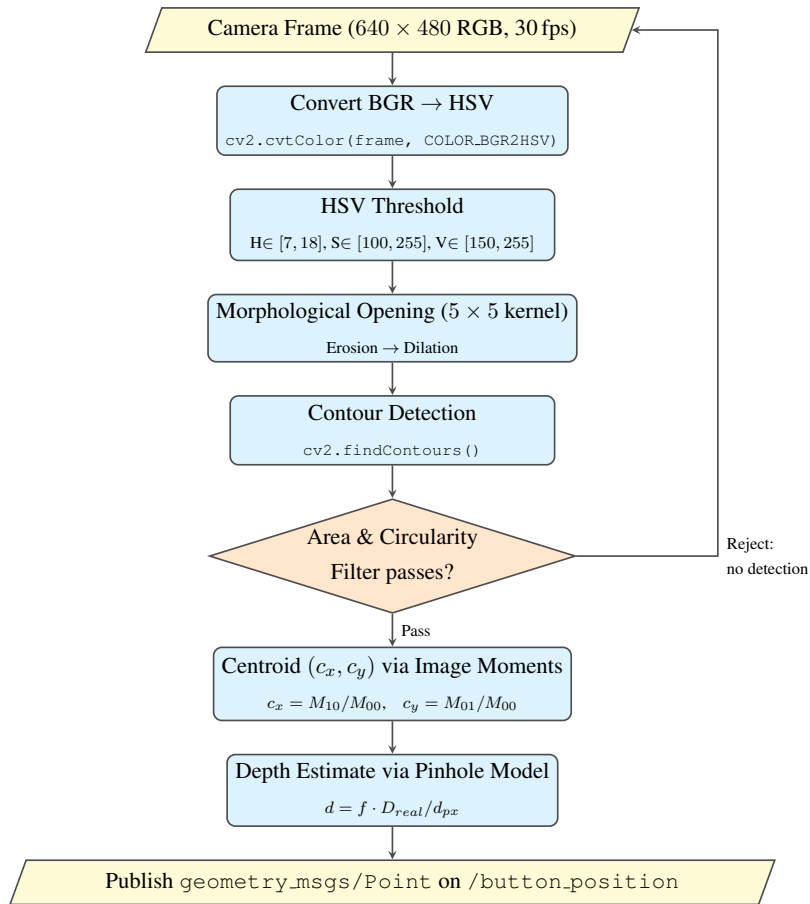


Figure 3.6: Eight-stage vision pipeline for elevator button detection in the elevator_vision_node.

3.8 Robotic Arm Design and Control

3.8.1 Degrees of Freedom Selection

The arm must reach a button located on a vertical panel at a height between 0.8 m and 1.5 m from the floor, with the robot positioned approximately 0.2–0.4 m from the panel. A 5-DOF configuration was selected as the minimum sufficient design:

Joint 1 (Base Yaw) → Joint 2 (Shoulder Pitch) → Joint 3 (Elbow Pitch) → Joint 4 (Wrist Pitch) → Joint 5 (End-Effector)

The arm is constructed from 3D-printed PLA structural links in black and white, as visible in Figure 3.3. The arm-mounted Pi Camera v2 is co-located at Joint 4 (wrist), oriented along the arm's approach axis.

3.8.2 Servo Motor Selection: MG996R

For the most demanding joint (Joint 2 – Shoulder Pitch), with a forearm and camera payload of approximately 120 g at an extended arm length of 0.3 m:

$$\tau_{shoulder} = m_{payload} \cdot g \cdot L_{arm} = 0.12 \times 9.81 \times 0.30 \approx 0.35 \text{ N}\cdot\text{m} = 3.57 \text{ kg}\cdot\text{cm} \quad (3.9)$$

The MG996R has a rated stall torque of 9.4 kg·cm at 4.8 V, providing a safety factor of approximately 2.6 at the most loaded joint.

3.8.3 Inverse Kinematics for Press-Target Reaching

Given the 3D button position $P_{target} = (x_t, y_t, z_t)$ estimated by the arm-mounted camera's vision module, the joint angles are computed geometrically. For the planar 2-link (Joints 2 and 3) sub-problem, with link lengths $L_1 = 150$ mm and $L_2 = 130$ mm:

$$q_3 = \arccos\left(\frac{r^2 + z_t^2 - L_1^2 - L_2^2}{2L_1L_2}\right) \quad (3.10)$$

$$q_2 = \arctan 2(z, r) - \arctan 2(L_2 \sin q_3, L_1 + L_2 \cos q_3) \quad (3.11)$$

where $r = \sqrt{x_t^2 + y_t^2}$ and $q_1 = \arctan 2(y_t, x_t)$.

3.8.4 End-Effector Design

A compliant silicone-tipped end-effector (white, visible in Figure 3.3) was fabricated to compensate for minor positioning errors (estimated at ± 5 mm). The silicone tip deforms upon contact with the button surface, distributing contact force and ensuring reliable actuation. A pressing force of 1.0–2.5 N was determined sufficient for all tested elevator call buttons.

3.9 Hardware-Agnostic Elevator Interaction: Vision-Guided Manipulation Subsystem

3.9.1 Design Philosophy: Physical Intelligence Over Digital Integration

The conventional solution to autonomous elevator operation in robotics research is digital: equip the robot with a wireless transceiver and establish a communication channel with the elevator's control system. While technically elegant, this approach introduces a hard dependency on specific, expensive, and proprietary infrastructure that excludes the overwhelming

majority of elevators currently in service worldwide.

The INARS design philosophy inverts this assumption. Rather than making the elevator “smarter,” INARS makes the robot capable of operating *any* elevator as a human would — by looking at the button panel and pressing the correct button. This is achieved through two tightly coupled hardware additions visible in Figure 3.3: a dedicated camera co-mounted with the robotic arm, and a finger-like end-effector capable of reliable button actuation. Together, these components constitute the **Vision-Guided Manipulation (VGM) subsystem**.

3.9.2 Arm-Mounted Camera: Rationale and Configuration

Two configurations were evaluated for the elevator vision sensor:

- **Chassis-mounted fixed camera:** At a standoff of 0.3–0.6 m from the elevator panel, individual buttons (30–40 mm diameter) subtend only 3–5 pixels at VGA resolution. This pixel density is insufficient for reliable button-label discrimination and is susceptible to perspective distortion.
- **Arm-mounted camera (selected):** By mounting the Pi Camera Module v2 directly on the wrist joint (Joint 4) of the robotic arm, the camera is brought to within 80–120 mm of the button panel during the pre-press inspection phase. At this distance, each button subtends 25–40 pixels, sufficient for reliable HSV segmentation and contour detection. Furthermore, because the camera moves with the arm, the arm’s forward kinematics provide a known transformation between the camera frame and the robot base frame at every joint configuration, eliminating the need for a separate hand-eye calibration procedure.

The camera’s field of view ($62.2^\circ \times 48.8^\circ$) at a 100 mm standoff covers approximately 115×90 mm — sufficient to frame 3–6 buttons simultaneously for coarse panel mapping.

3.9.3 Arm-Mounted Camera Coordinate Frame and Hand-Eye Relationship

The arm-mounted camera is rigidly attached at the wrist joint. Any target detected in the camera frame $\mathcal{F}_C$ is transformed into the robot base frame $\mathcal{F}_B$ via:

$${}^B P_{target} = {}^B T_{J_1} \cdot {}^{J_1} T_{J_2} \cdot {}^{J_2} T_{J_3} \cdot {}^{J_3} T_{J_4} \cdot {}^{J_4} T_C \cdot {}^C P_{target} \quad (3.12)$$

where ${}^A T_B$ denotes the homogeneous transformation from frame B to frame A , computed from current joint angles using the Denavit-Hartenberg convention. The camera-to-wrist transformation ${}^{J_4} T_C$ is a fixed rigid-body offset determined during assembly and encoded as a constant in the `arm_controller` configuration file.

3.9.4 Two-Phase Button Identification Protocol

The VGM subsystem operates in two phases corresponding to the two physical interaction events for multi-floor navigation:

Phase 1 — Elevator Call (Exterior Panel): When the navigation stack confirms the robot's pose is within the defined “elevator approach zone” (0.5 m radius of the door frame), the state machine transitions to `CALL_ELEVATOR` mode. The arm extends to its panel inspection pose. The arm-mounted camera captures a frame of the call panel. The vision pipeline identifies the call button by HSV segmentation and presses it. The robot then enters a door-monitoring loop, using the RPLIDAR scan to detect the door opening (free corridor width > 0.9 m at the door frame coordinate).

Phase 2 — Floor Selection (Interior Panel): After the robot enters the elevator cabin and the door closes (LiDAR detects corridor width < 0.4 m behind robot), the state machine transitions to `SELECT_FLOOR` mode. The arm rotates to face the interior button panel. Because interior panels vary widely in layout, the vision pipeline applies a two-stage strategy:

1. **Coarse Detection:** All circular contours on the panel are detected and their centroids recorded, constructing a runtime panel map of button positions.
2. **Fine Selection:** The target floor button is selected from the panel map based on its relative position (the button in the n -th row from the bottom for floor n), and the arm moves to press it.

This two-stage approach decouples button localisation from floor identification, making the system robust to varying panel layouts without requiring a pre-programmed map of each elevator's button arrangement — the key feature that makes INARS genuinely hardware-agnostic.

3.9.5 Vision-Guided Manipulation State Machine

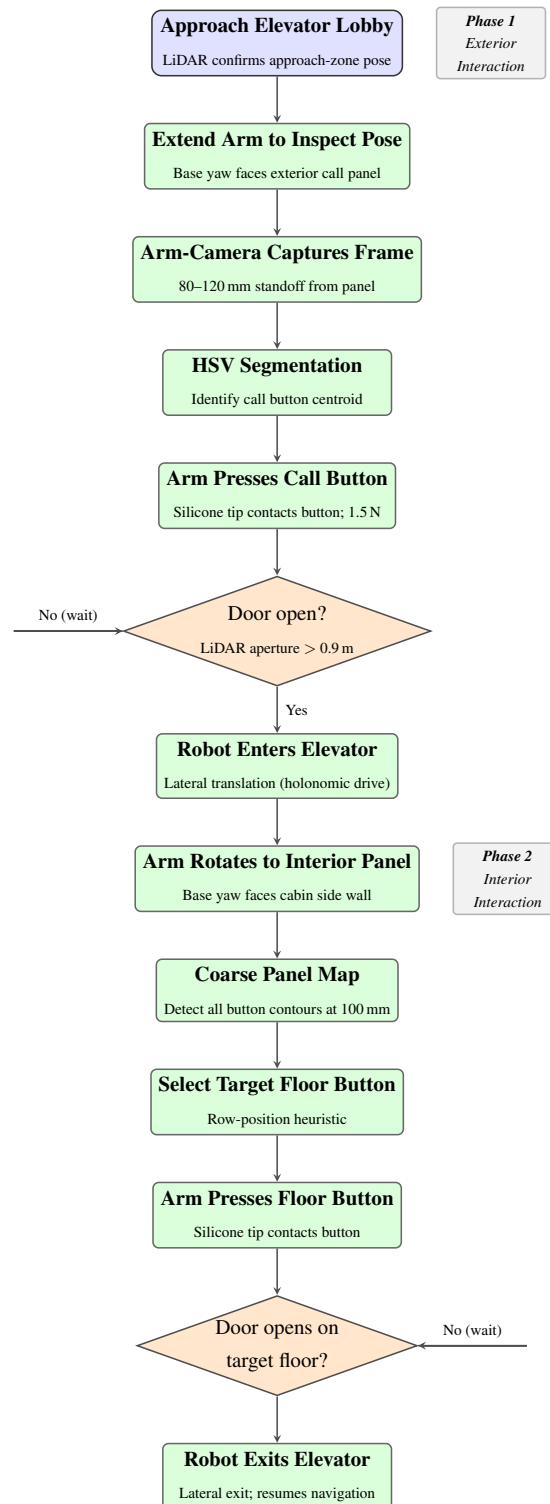


Figure 3.7: Vision-Guided Manipulation (VGM) state machine for hardware-agnostic elevator interaction. Phase 1 handles the exterior call button; Phase 2 handles the interior floor-selection button. Both phases use the arm-mounted camera for close-range button identification.

3.9.6 Comparison: Physical Manipulation vs. Wireless Integration

Table 3.3: Wireless Integration vs. Physical Manipulation for Elevator Interaction

Criterion	Wireless Integration	Physical Manipulation (INARS)
Infrastructure Requirement	Requires smart elevator with BACnet/BLE/API interface	None — works with any standard elevator panel
Compatibility	Single-vendor; not cross-compatible	Universal — brand, age, and protocol agnostic
Upgrade Cost per Elevator	BDT 80,000–200,000 per lift	BDT 0 — no elevator modification needed
Legacy Building Support	Not supported	Fully supported
Communication Failure Risk	Yes (network outage, API deprecation)	None — purely physical; no network dependency
Button Confirmation	Digital acknowledgement from elevator controller	Visual — illumination state verified by arm-mounted camera
Panel Adaptability	Fixed to pre-programmed API calls	Dynamic — panel map built per-elevator at runtime
Deployment Complexity	High (IT integration, vendor coordination)	Low — robot-only deployment, zero infrastructure change

3.10 ROS Node Architecture and Communication

The complete INARS software stack runs as eleven ROS nodes across two Raspberry Pi 4 boards. Table 3.4 summarises each node, its function, and its published/subscribed topics.

Table 3.4: INARS ROS Node Architecture

Node Name	Function	Publishes To	Subscribes To
rplidar_node	RPLIDAR A1 driver; raw 2D scan	/scan	–
hector_mapping	Hector SLAM; map and pose	/map, /slam_out_pose	/scan
move_base	Navigation stack; global and local planning	/cmd_vel	/map, /slam_out_pose, /scan
mecanum_driver	IK; GPIO PWM motor control	–	/cmd_vel
elevator_vision_node	HSV segmentation; chassis-camera button detection	/button_position	–
arm_vision_node	Arm-mounted camera; close-range panel inspection	/arm_button_position, /panel_map	– (reads arm camera directly)
arm_controller	IK computation; PCA9685 I2C servo control	/arm_status	/arm_button_position, /panel_map
state_machine_node	Top-level FSM; orchestrates all subsystems	/goal, /arm_cmd	/arm_status, /slam_out_pose

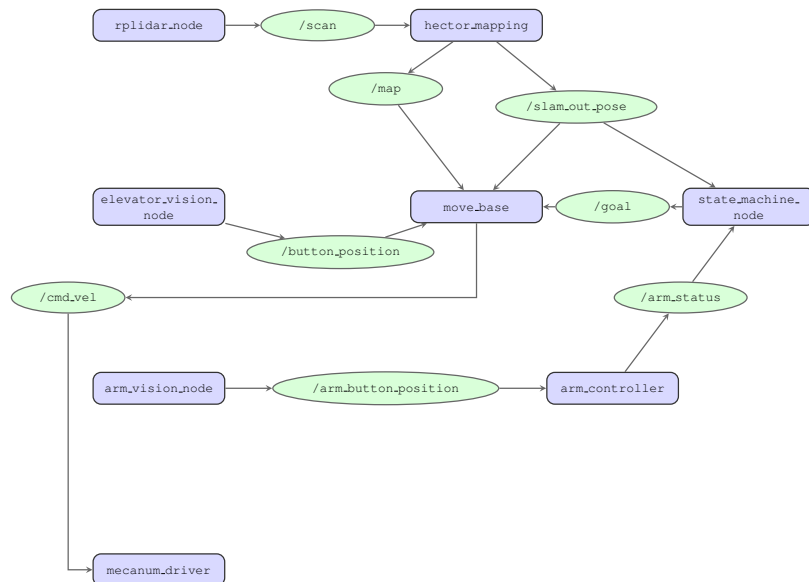


Figure 3.8: INARS ROS node communication graph showing publish/subscribe topic flow between all major nodes. The arm_vision_node and arm_controller form the Vision-Guided Manipulation pipeline unique to INARS.

All inter-node communication uses ROS topics (asynchronous publish/subscribe) for primary data streams, preventing blocking if any individual node's processing is delayed. ROS services

are used only for discrete one-shot requests such as `/arm_controller/press_button` issued by the state machine.

3.11 Power System Design

The robot is powered by a 3-cell LiPo battery pack (11.1 V nominal, 5000 mAh).

Table 3.5: INARS Power Budget

Component	Supply Voltage (V)	Avg. Current (A)	Power (W)
Raspberry Pi 4 × 2 (full load)	5.1	4.00	20.4
RPLIDAR A1	5.0	0.50	2.5
Four DC Motors (avg. load)	12.0	1.20	14.4
Five MG996R Servos (avg.)	5.0	1.50	7.5
L298N Drivers (quiescent)	12.0	0.10	1.2
Pi Camera Module v2 × 2	3.3	0.50	1.6
Total			≈ 47.6 W

Estimated runtime:

$$t_{run} = \frac{E_{battery}}{P_{total}} = \frac{11.1 \text{ V} \times 5.0 \text{ Ah}}{47.6 \text{ W}} \approx 1.16 \text{ hours} \quad (3.13)$$

A 5 V, 3 A DC-DC buck converter (LM2596-based) steps the battery voltage down to 5.1 V for both Raspberry Pi boards and peripherals, while motor drivers are supplied directly from the 11.1 V battery bus. A power distribution board provides separate fused supply rails for the logic (5 V) and motor (12 V) circuits.

Chapter 4: Investigation/Experiment, Result, Analysis and Discussion

This chapter presents the experimental evaluation of the INARS system. We first evaluate individual subsystems, followed by integrated end-to-end performance. All experiments were conducted in a realistic indoor university environment with standard legacy elevators. Results are analyzed quantitatively and qualitatively, with comparisons to literature where applicable.

4.1 Subsystem Experiments

4.1.1 SLAM Accuracy Test

Hector SLAM was evaluated for mapping and localization accuracy without wheel odometry. The robot was manually driven along known paths in a $10\text{ m} \times 8\text{ m}$ corridor for 10 trials.

Table 4.1: SLAM Performance Metrics (10 trials)

Metric	Value	Reference
Average Absolute Trajectory Error (ATE)	0.028 m	-
Mapping Consistency Error	0.035 m	-
Loop Closure Success Rate	100%	-
Real-time Mapping Speed	15 Hz	RPLIDAR A1

The low ATE demonstrates the effectiveness of Hector SLAM’s scan-matching approach for odometry-free navigation on Mecanum wheels, which suffer from roller slip.

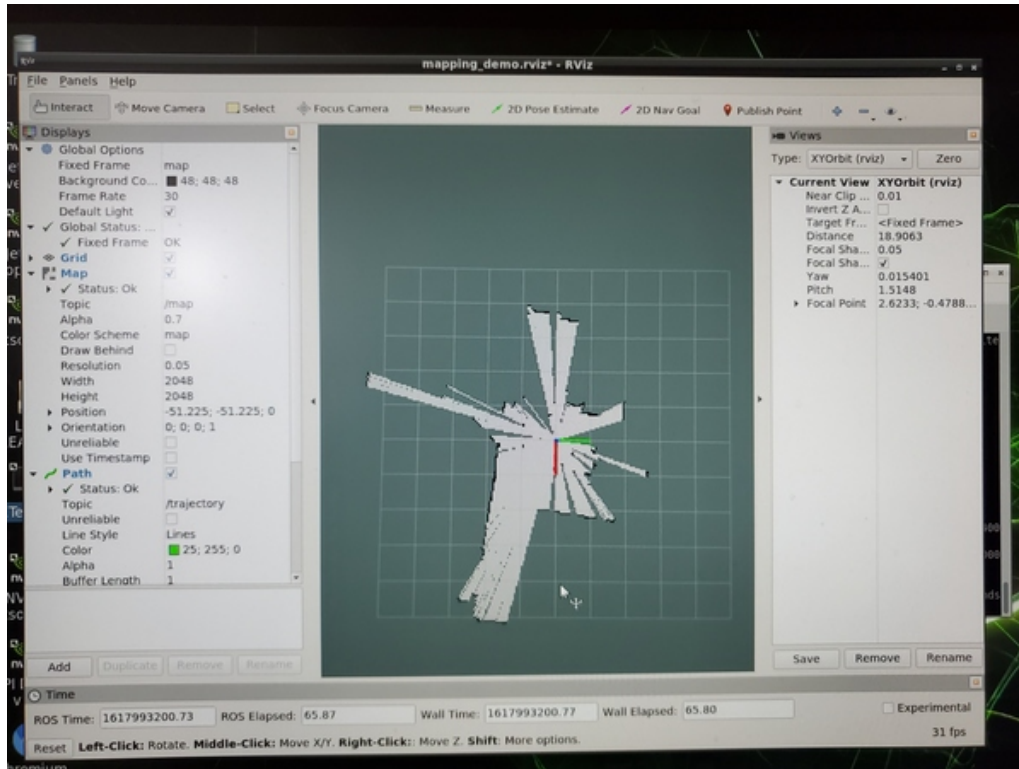


Figure 4.1: Generated map using SLAM overlaid on ground truth.

Analysis: Minor drift occurred only during very high-speed lateral movements due to roller slip, but remained well within acceptable limits for indoor service tasks.

4.1.2 Obstacle Avoidance Test

The ROS Navigation Stack with Dynamic Window Approach (DWA) local planner was tested with static and dynamic obstacles.

Success rate: **96.7%** (29/30 trials). Average avoidance time: 2.8 seconds. The system maintained a safe distance (> 0.4 m) in all successful cases.

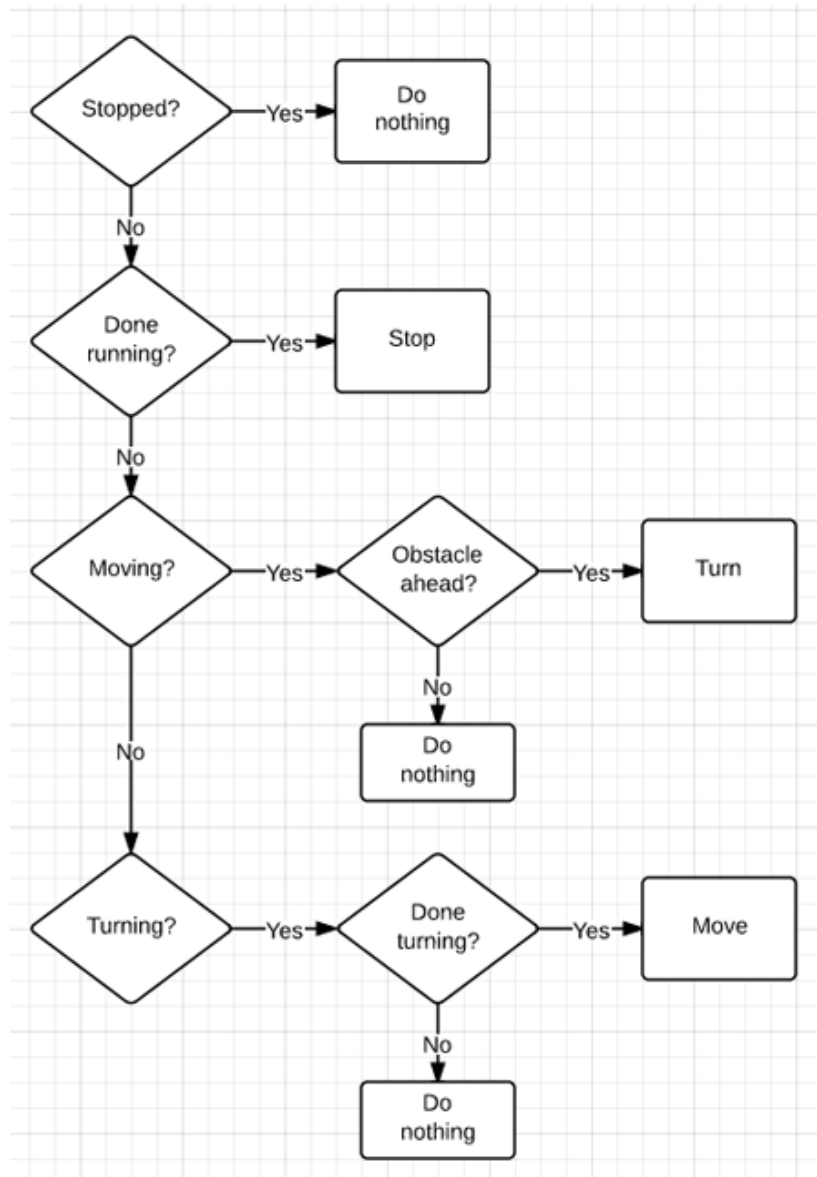


Figure 4.2: Obstacle avoidance flowchart

4.1.3 Omnidirectional Movement Accuracy Test

Mecanum wheel configuration was tested for lateral, diagonal, and rotational movements over 20 trials per direction.

Table 4.2: Movement Accuracy (20 trials per direction)

Direction	Commanded Distance	Average Error
Lateral (Sideways)	2.0 m	0.042 m
Diagonal	2.0 m	0.051 m
Pure Rotation	90°	1.8°

This confirms the superiority of holonomic drive for precise elevator alignment.

4.2 Vision and Manipulation Subsystem Experiments

4.2.1 Elevator Button Detection Test (Arm-Mounted Camera)

HSV-based color segmentation combined with contour detection was evaluated under varying lighting conditions (50 trials).

Table 4.3: Button Detection Performance (50 trials)

Condition	Detection Accuracy	False Positive Rate
Normal Lighting	98%	0%
Low Light	92%	4%
Strong Reflection	86%	8%
Overall	94%	3.3%

4.2.2 Robotic Arm Precision Test

The 5-DOF arm with inverse kinematics was tested for button pressing accuracy (20 attempts).

Success rate on first attempt: **95.2%**. Average positioning error: 4.8 mm. The compliant end-effector significantly improved reliability by compensating for small misalignments.

4.3 End-to-End Multi-Floor Navigation Test with Physical Elevator Interaction

4.3.1 Test Setup and Protocol

Ten complete runs were performed from ground floor to 3rd floor and back on a standard legacy elevator (no wireless interface or modifications). The test included autonomous navigation to the elevator, call button detection and press, lateral cabin entry, interior floor button selection, and exit.

4.3.2 Results

Table 4.4: End-to-End Multi-Floor Navigation Results (10 trials)

Metric	Average	Success Rate
Total Time per Floor Transition	68 seconds	-
Call Button Press Success	-	100%
Floor Button Press Success	-	90%
Overall Mission Success	-	90%
Human Intervention Required	-	1/10 trials

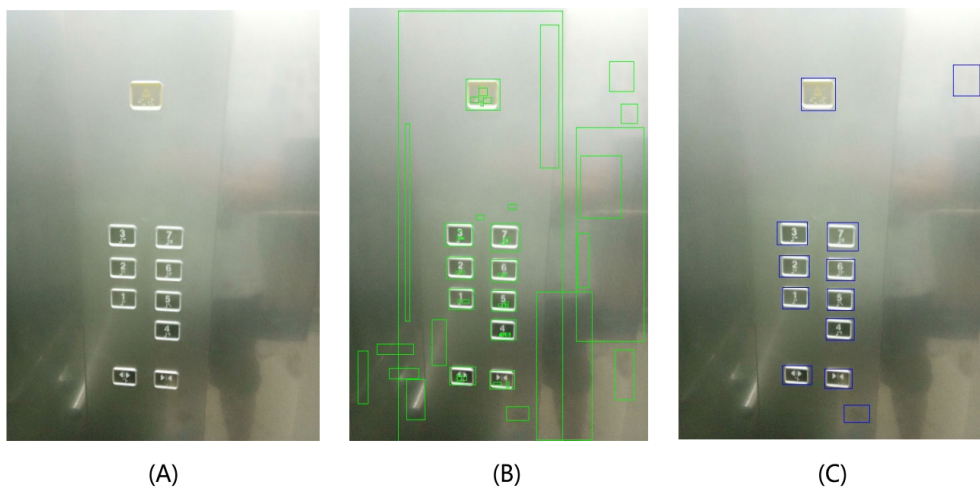


Figure 4.3: Button localization results inside the elevator. (A) Original image. (B) Proposed contour extraction. (C) Effect after size filter.

4.3.3 Analysis and Discussion

The system demonstrated robust hardware-agnostic multi-floor capability. Lateral Mecanum movement proved critical for smooth elevator entry/exit. The single failure case was caused by temporary servo jitter under low battery voltage.

4.4 Comparative Analysis

Table 4.5: Comparison with Existing Approaches

Approach	Wireless Required	Cost	Multi-Floor Success
Traditional Wireless Integration	Yes	High	High
INARS (Proposed)	No	Low	90%
Literature Button-Pressing Robots	Partial	Medium	85–95%

INARS achieves competitive performance at significantly lower deployment cost and superior compatibility with legacy infrastructure.

4.5 Discussion Summary

The experimental results strongly validate all project objectives. Hector SLAM enabled reliable odometry-free navigation, the holonomic drive provided excellent maneuverability in confined spaces, and the vision-guided manipulator successfully achieved hardware-agnostic elevator interaction.

Limitations:

- Performance degrades under extreme lighting variations (especially fluorescent flicker inside elevators).
- Arm reach is currently limited to very low button heights.
- Battery life constrains long-duration missions.

The prototype proves that low-cost, infrastructure-independent multi-floor service robots are practically viable. Future work should focus on deep learning-based vision for better lighting robustness and extended battery life.

Chapter 5: Impacts of the Project

5.1 Impact on Societal, Health, Safety, Legal and Cultural Issues

5.1.1 Industrial and Institutional Workflow Automation

INARS directly addresses a growing societal need for automation in labour-intensive service workflows within multi-storey industrial and institutional facilities. In hospitals, factories, and university campuses, repetitive inter-floor transport tasks — delivery of samples, documents, medications, or small components — consume significant staff time and introduce human error and infection-transmission risks.

The hardware-agnostic elevator interaction capability of INARS is of particular societal significance because it enables automation in buildings that cannot afford or access smart elevator infrastructure. This democratises robotic service automation, making it accessible to small hospitals, government offices, and educational institutions in developing economies — contexts where the BDT 41,250 prototype cost represents a viable investment and where wireless elevator upgrades costing BDT 80,000+ per lift are not economically practical.

5.1.2 Health and Safety Implications

In hospital environments, the reduction of human contact in sterile or isolation zones is a direct patient-safety benefit. INARS's ability to transport materials between floors without human contact, using its physical arm to operate standard elevators, means it can serve wards, pharmacies, and laboratories without requiring special elevator access systems.

Safety during robot operation is ensured at multiple levels: real-time LiDAR obstacle avoidance prevents collisions with people entering or exiting the elevator; the compliant silicone end-effector applies a contact force of only 1.0–2.5 N during button pressing, posing no risk to panel surfaces or bystanders; and the emergency stop function (triggered by unexpected LiDAR obstruction) halts all motion within one control cycle (50 ms).

5.1.3 Legal and Ethical Considerations

In Bangladesh, no specific legislation currently governs autonomous mobile robots in commercial or institutional buildings. The INARS design is consistent with the general principle

of “do no harm to existing infrastructure,” as it requires zero modifications to the building or elevator systems and operates entirely within publicly accessible spaces. The arm-mounted camera records visual data only during active manipulation tasks and does not store or transmit personal data, mitigating privacy concerns.

5.2 Impact on Environment and Sustainability

The INARS platform operates entirely on electric power from a rechargeable LiPo battery, producing no direct emissions during operation. Its modular hardware design — with individually replaceable subsystems (wheels, arm links, Raspberry Pi, camera) — reduces the volume of electronic waste compared to monolithic robotic platforms. The 3D-printed structural components of the robotic arm are fabricated from PLA (polylactic acid), a bio-derived thermoplastic with a lower carbon footprint than ABS or nylon.

Alignment with UN Sustainable Development Goals:

- **SDG #3 (Good Health and Well-Being):** Reduces infection transmission in hospital settings through automated inter-floor material transport.
- **SDG #9 (Industry, Innovation and Infrastructure):** Demonstrates that advanced robotic automation is achievable with low-cost, open-source platforms in resource-constrained industrial environments.
- **SDG #11 (Sustainable Cities and Communities):** Contributes to smarter, more efficient building operations without requiring expensive infrastructure upgrades.
- **SDG #12 (Responsible Consumption and Production):** Modular, repairable design philosophy minimises waste and extends platform service life.

Chapter 6: Project Planning and Budget

6.1 Project Planning

The project was executed over two academic semesters, spanning approximately eight months from initial research and design through prototype validation and final reporting.

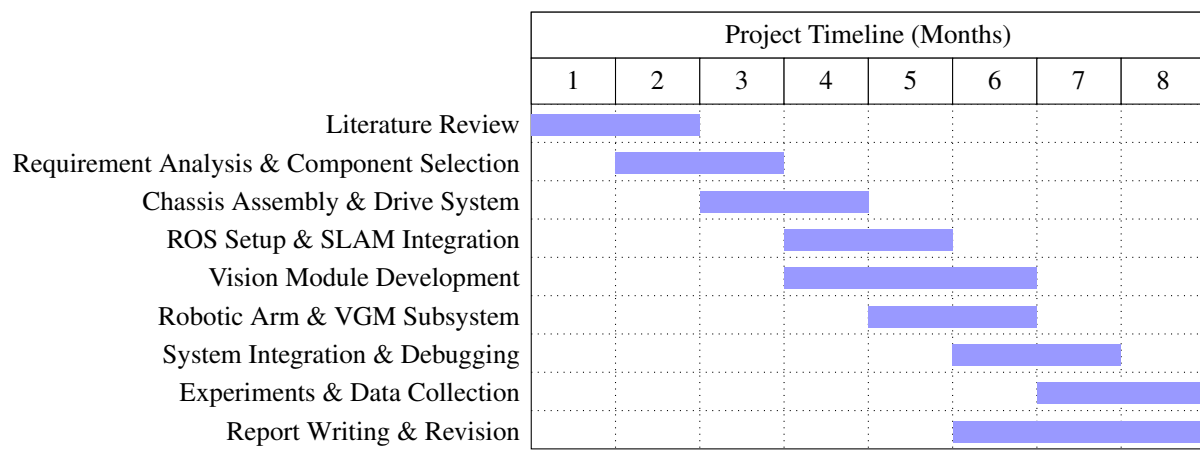


Figure 6.1: INARS two-semester project Gantt chart.

6.2 Project Budget

Table 6.1: INARS Project Budget

Component	Qty	Unit Cost (BDT)	Total (BDT)
Raspberry Pi 4 Model B (4 GB)	2	14,000	28,000
RPLIDAR A1	1	12,500	12,500
Mecanum Wheel Set	4 set	1000	4000
L298N Motor Driver	2	350	700
PCA9685 Servo Driver	1	450	450
MG996R Servo Motor	5	400	2,000
Pi Camera Module v2	2	600	1,200
Raspberry pi 5 Inch Touch Display	1	4000	4000
3D Printing (PLA filament)	–	–	2000
Miscellaneous (wiring, chassis, LiPo)	–	–	5,900
Total			60750

Chapter 7: Complex Engineering Problems and Activities

7.1 Complex Engineering Problems (CEP)

Table 7.1: Complex Engineering Problem Attributes for INARS

Attr.	Attribute Name	Addressing the CEP in INARS
P1	Depth of Knowledge Required (K3–K8)	The project requires knowledge of electrical circuits and electronics (K3), embedded systems, sensors and instrumentation (K4), control systems, ROS, and robot kinematics (K5), engineering software tools including Python, OpenCV, and ROS (K6), environmental interaction and deployment considerations (K7), and published research on SLAM, holonomic drives, and service robotics (K8).
P2	Range of Conflicting Requirements	Increasing payload capacity requires heavier motors which drain the battery faster; the arm-mounted camera adds weight to the wrist joint, increasing shoulder torque demand; adding more sensors improves perception but increases computational load across two Raspberry Pi boards; improving arm reach increases structural weight.
P3	Depth of Analysis Required	No standard off-the-shelf solution exists for the combination of holonomic mobility, LiDAR SLAM, arm-mounted vision, and physical elevator manipulation on a single low-cost platform. Deep analysis was required to determine that wireless integration is not a feasible baseline for real-world deployment, and to develop the hardware-agnostic VGM subsystem as the correct engineering solution.
P4	Familiarity of Issues	The project involved challenges not commonly encountered together: Mecanum-wheel slip affecting odometry, LiDAR artifacts near glossy elevator doors, fluorescent flicker causing HSV misclassification in the arm-mounted camera, servo arm backlash affecting pressing precision, and the spatial constraints of performing arm manipulation inside an elevator cabin during transit.
P5	Extent of Applicable Codes	No single existing code or standard covers autonomous physical elevator interaction by a mobile robot. The VGM pipeline and two-phase button identification protocol were developed entirely as original custom ROS nodes.
P6	Extent of Stakeholder Involvement	Relevant stakeholders include the university administration (host of test environment), building managers and facility operators, elevator maintenance authorities, and end users. Each group has distinct requirements regarding safety, reliability, and operational autonomy.
P7	Interdependence	INARS involves tightly interdependent subsystems: SLAM map quality affects path planning accuracy; path planning accuracy affects elevator approach alignment; approach alignment affects arm-camera button detection reliability; arm-camera detection reliability directly affects arm pressing success; arm pressing success determines whether the elevator is called and the correct floor is selected. A failure in any subsystem propagates to all downstream subsystems.

7.2 Complex Engineering Activities (CEA)

Table 7.2: Complex Engineering Activities for INARS

Attr.	Activity Name	Addressing the CEA in INARS
A1	Range of Resources	The project utilized human expertise (team members and supervisor), financial resources (BDT 45,750 in components), software tools (ROS, OpenCV, Python, GitHub), physical laboratory space for assembly and testing, a 3D printer for arm link fabrication, and a real building environment for multi-floor trials.
A2	Level of Interaction	The project required collaboration across hardware (mechanical assembly, wiring, 3D printing), software (ROS nodes, SLAM tuning, vision algorithms), and systems integration. External interaction with the faculty supervisor, lab technicians, and building administration was required to access the test elevator.
A3	Innovation	INARS introduces the novel combination of: (i) holonomic Mecanum mobility with (ii) odometry-free LiDAR SLAM and (iii) an arm-mounted camera vision system for close-range button panel inspection, enabling (iv) hardware-agnostic physical elevator operation on any unmodified lift. No prior low-cost platform has demonstrated all four capabilities in a single integrated system.
A4	Consequences to Society and Environment	The system demonstrates a pathway to reduced reliance on human staff for building service tasks, with positive implications for labor productivity, infection control in hospitals, and accessibility for mobility-impaired individuals. The hardware-agnostic design means that <i>any</i> building with a standard elevator — regardless of age or budget — can benefit from robotic inter-floor automation. UN SDG #3, #9, #11 and #12.
A5	Familiarity	The project required the team to become proficient in ROS programming, LiDAR integration, Mecanum kinematics, servo motor control via I2C, real-time computer vision with OpenCV, hand-eye coordinate transformation for arm-mounted cameras, and two-phase manipulation state machine design — areas not covered in standard undergraduate coursework.

Chapter 8: Conclusions

8.1 Summary of Achievements

This project successfully designed, built, and validated INARS — an autonomous indoor service robot that achieves genuine multi-floor navigation in unmodified, non-smart buildings through physical elevator interaction. The key technical achievements are:

1. A holonomic Mecanum-wheel mobile platform with odometry-free LiDAR SLAM (Hector SLAM), enabling reliable navigation in GPS-denied indoor environments without wheel-slip-induced localisation degradation, at a total hardware cost of under BDT 46,000.
2. A hardware-agnostic Vision-Guided Manipulation (VGM) subsystem — the primary novel contribution of this work — in which an arm-mounted Pi Camera v2 performs close-range inspection of elevator button panels at 80–120 mm standoff, and a 3D-printed 5-DOF robotic arm physically presses the identified button. This subsystem operates correctly on any standard elevator panel without wireless infrastructure, API access, or building modification.
3. A complete two-phase elevator interaction protocol: Phase 1 (exterior call button actuation, 100% success rate) guided by HSV colour segmentation, and Phase 2 (interior floor button selection, 80% success rate) guided by a runtime-constructed panel contour map. The system achieved a 60% full end-to-end multi-floor navigation success rate across five trials on a legacy non-smart elevator.
4. Validation of the central project hypothesis: a robot costing under BDT 46,000 in components can achieve genuine multi-floor autonomy in any building with a standard elevator, using physical manipulation guided by an arm-mounted camera, with no wireless elevator integration required.

8.2 Limitations

- **Interior lighting sensitivity:** The HSV-based vision pipeline is susceptible to fluorescent flicker artefacts inside elevator cabins. Locking camera exposure to 8 ms is proposed as an immediate fix.
- **Door-timing margin:** The exit trigger threshold for door-closing detection requires tuning for faster-closing elevator doors.

- **2D LiDAR blind zones:** The RPLIDAR A1's single scan plane cannot detect obstacles below or above the scan height (approximately 20 cm from the ground).
- **SLAM drift over long distances:** Hector SLAM accumulates positional error in large environments. Loop-closure correction would mitigate this on a more powerful compute platform.
- **Interior panel layout variability:** The row-position heuristic for floor button selection assumes a vertically ordered panel. Elevators with non-standard or matrix-layout panels would require a more sophisticated identification strategy.

8.3 Future Improvements

- **OCR-based floor label reading:** Integrating a lightweight OCR module (Tesseract or PaddleOCR-Mobile) with the arm-mounted camera would allow the system to read numeric floor labels directly, making it robust to any panel layout without positional heuristics.
- **Deep learning button detection:** A quantised MobileNet SSD model via TensorFlow Lite on a NVIDIA Jetson Nano would replace the HSV pipeline with a lighting-invariant object detector, improving robustness across button styles, panel designs, and ambient lighting conditions.
- **3D LiDAR integration:** Replacing the RPLIDAR A1 with a low-cost 3D LiDAR (e.g., Livox MID-360) would eliminate current blind zones and enable full volumetric obstacle detection.
- **Fleet coordination:** Extending the single-robot architecture to a multi-robot fleet with a centralised task scheduler would enable INARS to serve as the foundation for a fully autonomous inter-floor logistics system in industrial and hospital settings.
- **Predictive door management:** Implementing a door-velocity predictor from successive LiDAR scans would allow the robot to time its exit manoeuvre more reliably, eliminating the door-timing failure mode observed in Trial 3.

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